

| Question Number | Answer                                                                                                                                                                                                                                                                                            | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>2(a)(i)</b>  | <p>The only correct answer is C 0.48</p> <p><i>A is not correct because <math>2 \times 0.6 \times 0.4 = 0.48</math></i></p> <p><i>B is not correct because <math>2 \times 0.6 \times 0.4 = 0.48</math></i></p> <p><i>D is not correct because <math>2 \times 0.6 \times 0.4 = 0.48</math></i></p> | <b>(1)</b> |

| Question Number | Answer                                                             | Additional guidance              | Mark       |
|-----------------|--------------------------------------------------------------------|----------------------------------|------------|
| <b>2(a)(ii)</b> | <ul style="list-style-type: none"> <li><math>q^2</math></li> </ul> | <p>IGNORE q</p> <p>IGNORE q2</p> | <b>(1)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                               | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>2(b)</b>     | <p>An explanation that makes reference to four of the following points:</p> <ul style="list-style-type: none"> <li>• genetic variation in Lokrum island rabbit population (1)</li> <li>• viral disease acts as selection pressure (1)</li> <li>• some rabbits had advantageous allele(s) that enabled them to survive (the viral disease) (1)</li> <li>• (these) rabbits reproduced and passed the advantageous allele(s) on to their offspring (1)</li> <li>• increasing the frequency of the {advantageous / beneficial} allele(s) (1)</li> <li>• one gene may have greater effect so there will be a greater change in allele frequencies for this gene (1)</li> </ul> | <p>ACCEPT rabbits have different alleles<br/>ACCEPT mutation resulted in {genetic variation / different allele / advantageous allele}</p> <p>IGNORE gene / immune / characteristics<br/>ACCEPT converse</p> <p>IGNORE gene</p> <p>IGNORE gene</p> | <b>(4)</b> |